

Sakhi Bazaar

AI-Enabled Marketplace for Women Entrepreneurs Technical Specifications & Development Roadmap

Document Purpose & Context

This document details the functional requirement modules and implementation milestones for the Sakhi Bazaar platform. It serves as the companion specification sheet to the project review presentation, providing comprehensive tracking of completed features and upcoming database, administrative, and payment integration pipelines.

Platform Version:	v1.0 (MVP Completed, Phase 2 Active)
Architecture Stack:	MERN (MongoDB, Express, React 19, Node.js)
Integrations:	Google Gemini API, Firebase Authentication, Cloudinary CDN
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1. Functional Requirement Modules

Below are the structured modules representing the complete functional footprint of Sakhi Bazaar. Each requirement maps to a specific user role and indicates current implementation status.

Module 1: Authentication & Onboarding

Req ID	Description	Role	Status
FR-1.1	Email/Password Registration and Login	All	Implemented
FR-1.2	Google Authentication via Firebase SSO	All	Implemented
FR-1.3	Role Assignment (customer vs. seller at signup)	All	Implemented
FR-1.4	JWT Generation & HTTPOnly Cookie Token Management	All	Implemented
FR-1.5	Seller Profile Completion (Store details, contact info)	Seller	Planned

Module 2: Buyer Storefront & Browsing

Req ID	Description	Role	Status
FR-2.1	Public landing page displaying recently added products	Buyer	Implemented
FR-2.2	Real-time text search (searches title & description)	Buyer	Implemented
FR-2.3	Category-based filters (Clothing, Handmade, Food)	Buyer	Implemented
FR-2.4	Detailed product view page with seller contact information	Buyer	Implemented
FR-2.5	Shopping cart (add, remove, adjust quantities)	Buyer	Planned

Module 3: Seller Product Management

Req ID	Description	Role	Status
FR-3.1	List a new product (title, price, category, images)	Seller	Implemented
FR-3.2	Secure product image uploads to Cloudinary storage	Seller	Implemented
FR-3.3	Update existing product details and change images	Seller	Implemented
FR-3.4	Delete listed products (removes from marketplace)	Seller	Implemented
FR-3.5	Seller Dashboard listing only products created by seller	Seller	Implemented

Module 4: Generative AI Assistant (Gemini API)

Req ID	Description	Role	Status
FR-4.1	AI Description: Storytelling copy writing from titles	Seller	Implemented
FR-4.2	AI Caption: Emoji-rich captions for social promotion	Seller	Implemented
FR-4.3	One-click copy to clipboard functionality	Seller	Implemented

Module 5: Market Price Awareness

Req ID	Description	Role	Status
FR-5.1	Local Platform Metrics: Min, max, and avg category pricing	Seller	Planned
FR-5.2	AI Valuation: Gemini recommendations during entry	Seller	Planned
FR-5.3	Price Comparison Widget for customer transparency	Buyer	Planned

Module 6: Payment Gateway Integration

Req ID	Description	Role	Status
FR-6.1	Checkout System: Initiate purchase request from cart	Buyer	Planned
FR-6.2	Stripe/Razorpay API Integration: Credit/debit/UPI payments	Buyer	Planned
FR-6.3	Webhook Handler: Verify transactions on the backend	System	Planned
FR-6.4	Payment Confirmation: Digital receipts & status updates	Buyer/Seller	Planned

Module 7: Administrative Portal

Req ID	Description	Role	Status
FR-7.1	Admin Role Access: Dedicated database administrator key	Admin	Planned
FR-7.2	Vetting Dashboard: Review and approve business credentials	Admin	Planned
FR-7.3	Product Moderation: Suspend/flag inappropriate listings	Admin	Planned
FR-7.4	Financial Metrics: Track platform sales volumes & commission	Admin	Planned

2. Development Roadmap & Tracking Checklist

Phase 1: MERN Foundation & AI Integration (MVP) - COMPLETED

- ✓ Initialize backend Node/Express server and setup Mongoose models.
- ✓ Design initial MongoDB schemas for User, Product, and Chat logs.
- ✓ Implement JWT Authentication and bcrypt password encryption.
- ✓ Design responsive public web layouts using Tailwind CSS.
- ✓ Implement buyer login, registration, and seller dashboards on React client.
- ✓ Connect Cloudinary upload middleware to secure product image uploads.
- ✓ Program backend Gemini routes for AI Description & Caption Generation.
- ✓ Integrate one-click copy-to-clipboard elements on merchant dashboard.

Phase 2: Market Price Awareness & Payments - ACTIVE DEVELOPMENT

- ☐ Create backend API endpoint `/api/products/stats/category` to fetch pricing metrics.
- ☐ Render price benchmark stats inside seller AddProduct/EditProduct dashboards.
- ☐ Integrate secure payment gateway checkout routes (`/api/payments/checkout`).
- ☐ Embed Stripe Elements or Razorpay Modal inside buyer checkout pages.
- ☐ Program secure webhook endpoints on backend to verify merchant success.

Phase 3: Administrative Control & Portal - UPCOMING

- ☐ Extend User database schema to support specific Admin security access.
- ☐ Secure backend admin API routes with validation middleware (`isAdmin`).
- ☐ Construct Admin Dashboard React panel to monitor global products and buyers.
- ☐ Add Deactivate Account and Flag Product moderation elements.
- ☐ Implement transaction graphs showing platform sales volume.